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A Put Option Paradox

Mark Grinblatt and Herb Johnson*

Abstract

What happens to the price of a put in a period during which the stock price stays constant? The hedging strategy implicit in the Black-Scholes model would seem to imply that the put goes up in value. Pure arbitrage arguments imply the opposite result. This paper resolves the paradox and uses it to explore the restrictions inherent in the diffusion processes assumed for all option pricing models.

I. Introduction

Many option pricing theorists are aware that the stochastic diffusion processes that permit hedges in option pricing models are restrictive. In particular, we know that random variables generated by these stochastic processes never stay still. In this paper, we will illustrate an interesting paradox that arises when one ignores this fact.

II. The Paradox

Consider a stock and an out-of-the-money finite-lived American (or sufficiently out-of-the-money European) put option on the stock. If the stock follows a diffusion process with known variance (e.g., geometric Brownian motion), a trader can form a self-financing, riskless hedge in continuous-time by buying puts and the underlying stock in the appropriate amounts. Now suppose that the stock price remains constant (over an infinitesimal or finite time period).¹ The hedge involves a positive investment in both securities and so must earn a positive riskless return regardless of what happens to the stock price. If the stock price does not change, then the put price must go up in order for the hedge to have a positive return. Equivalently, a put is perfectly replicated by a short position in the stock and a long position in a riskless, pure discount bond. If the stock price does not change in an infinitesimally small time interval, the portfolio of stocks and bonds, and hence the put, must go up in value.

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¹ Because option prices are continuous functions of stock prices, this paradox also arises for small negative movements in the stock price.

So long as the (European) put is (sufficiently) out-of-the-money, these arguments must be fallacious. They imply, for instance, that the put would have positive value at maturity if the stock price did not change in every infinitesimal time interval until the maturity date of the put. This is impossible since the put would be out-of-the-money at the maturity date if it were initially out-of-the-money. Moreover, if the put is an out-of-the-money American put, we know that its value would continually decline as maturity approached if its stock price did not change.² Finally, the closed form Black-Scholes (1973) solution for the European put suggests that for some parameter values, the put price decreases.

III. The Resolution

To see what is wrong, let us investigate the binomial process. Cox, Ross, and Rubinstein (1979) (see also (1985)) have shown that a binomial process, with a period defined to be sufficiently small, can approximate a continuous-time process to any degree of accuracy. That is, for periods of length $\Delta t = (T - t)/n$, letting

$$(1) \quad u = \exp(\sigma\sqrt{\Delta t}) \text{ and } d = \exp(-\sigma\sqrt{\Delta t}),$$

the limit of the binomial process with parameters u and d is geometric Brownian motion as n becomes infinite.

A stock price S that is governed by a binomial process can assume one of two values next period, uS and dS , where $u > 1 > d$. Note that it is uninteresting to set u or d equal to 1, and hence it makes little sense to ask what happens if the stock price does not change over one period.³ Of course, when $d = 1/u$, the stock price may be unchanged after two periods. However, in this case, the put-replicating portfolio of stock and riskless bonds gets “whipsawed” in the intermediate period, since it calls for selling more shares short as the stock price falls and buying some back (but still maintaining a net short position) when the stock price rises. The effect of selling low and buying high leads to a put value that declines over the two periods. *Ex post*, an investor regards the put-replicating portfolio to be a bad investment, given that the stock price did not change over two periods.

This intuition extends to the continuous-time stochastic processes. According to Equation (1), one-period returns are of the order of $\sqrt{\Delta t}$, which is larger than Δt . Thus, in any finite amount of time, Δt , the probability that the stock price takes on values other than its value at the beginning of the period is 1. This, however, is stronger than the usual statement that a random variable with a continuous distribution has a zero probability (and not a zero density) of taking on a particular value (such as its initial value). For instance, consider the interval

$$I_{\Delta t} = \{S: S_t(1 - \Delta t) < S < S_t(1 + \Delta t)\}.$$

² This is because an American option with a longer time to expiration contains all the rights of a shorter maturity option with otherwise identical features—plus more. Hence, to prevent arbitrage, the former option must sell for at least as much as the latter. With a geometric Brownian motion process for the stock, the former option must sell for more.

³ If $d = 1$, an out-of-the-money put option always stays out-of-the-money and is worth nothing. If $u = 1$, and both interest rates and the stock price are positive, there is arbitrage. Simply short the stock and purchase riskless bonds with the proceeds of the short sale.

There is some probability $\pi_{\Delta t}$ that the stock price S_τ will lie outside this interval for at least one point in time τ where $t < \tau < t + \Delta t$. However, as Δt approaches 0, $\pi_{\Delta t}$ approaches 1. This is because a fundamental property of geometric Brownian motion is that it does not allow one to resolve uncertainty by partitioning time into smaller and smaller intervals. Stated another way, although the density function of $S_{t+\Delta t}/S_t$ converges to a function with zero density on points other than 1 as Δt approaches 0, it converges at a rate much slower than Δt .

While mathematically imprecise, one can heuristically imagine that over intervals of length dt , the whipsaw effect described for the binomial process is still occurring with the geometric Brownian motion, at a profit-affecting level of the order of $\sqrt{dt}$. However, in contrast to the two-period binomial example, this whipsaw is occurring an infinite number of times rather than once.

One implication of the “paradox” is that attempts to continuously hedge are bound to fail for any nontrivial stochastic process that allows the stock price to remain absolutely constant over some finite interval. Any self-financing portfolio that is long in both puts and stocks cannot earn the risk-free rate if neither security earns a positive return, which would be true if the put declined in value. However, arbitrage considerations imply that sufficiently out-of-the-money European and nonexercised American puts must decline in value.

An alternative interpretation of this inability to hedge explores differences between the *ex ante* and *ex post* variances of the stock return. In the Black-Scholes model, hedge portfolio weights are based on *ex ante* variances. Hedging strategies succeed *ex post* because the *ex post* variance cannot differ from the *ex ante* variance over a finite time interval with the geometric Brownian motion process. With this process, the probability density function for the *ex post* variance at the end of a finite time interval is degenerate, with value 0 everywhere but at the *ex ante* variance. Note that this is a much stronger statement than saying that the probability of any particular future value of a continuous random variable is 0. Equality between *ex ante* and *ex post* variances need not hold, however, with other stochastic processes, such as a process that allows stock prices either to remain constant or to vary, certain jump processes, and processes that have changing *ex ante* variances. In these cases, there is generally no risk-free hedge with the option and the stock alone.

The stochastic processes that govern stock price paths in the real world may be processes that allow the *ex post* variance to differ from the *ex ante* variance over finite time intervals. If so, then researchers testing the efficiency of the options market and traders must be alert to this problem. In particular, researchers who seem to find abnormal returns from ostensibly riskless hedges should check to see if the returns were merely generated by a strategy that benefits from the *ex post* variance differing from the *ex ante*.

IV. Summary

This paper has presented and resolved a paradox concerning what happens to the price of a put when the stock price does not change.

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